SYSDESIGN REFERENCE

Message Queues

Core concepts, reference diagram, tradeoffs, and interview prompts.

Decouple producers from consumers and absorb bursts with explicit delivery semantics.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / MESSAGE-QUEUES

SYSDESIGN

Message Queues

CORE IDEA

Decouple producers from consumers and absorb bursts with explicit delivery semantics.

Reference architecture decision chain

01 Ordering	02 Partitions	03 Retention	04 Retries

Decisions to defend

+ Ordering
+ partitions
+ retention
+ retries
+ dead letters
+ consumer groups
+ and idempotency.

Failure modes to test

+ Poison messages
+ backlog growth
+ duplicate effects
+ and reordering.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Message Queues

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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